

Optical See-through Scope for Observing the Global Component of a Scene

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ABSTRACT

The reflected light on a physical object can be separated into a direct illumination component and a global illumination component. Observing only the global component provides a better understanding of the reflectance property of the surface. In conventional methods, the components are separated using computer vision techniques. In this study, we developed a see-through optical scope that allows a user to selectively observe the global component using high spatial frequency pattern projection and a liquid crystal panel. We confirmed that only the global component could be observed with the naked eye through the scope.

Index Terms: Human-centered computing—Visualization—Visualization application domains—Scientific visualization; Human-centered computing—Visualization—Visualization systems and tools—Visualization toolkits;

1 INTRODUCTION

The human visual system understands the shape and texture of a physical object by observing the reflected light of light sources, such as the sun or room lights, on the object. The observed reflected light can be separated into a direct illumination component and a global illumination component [2]. The direct component is the light reflected on a physical surface and enters the observer's eye without being reflected again on other surfaces. The global component is the other reflected light component, such as inter-reflection and subsurface scattering. Although the global component is considered to be a disturbance in display technologies [4], it also conveys rich information about the scene that is imperceptible for normal human vision. Researchers have applied technologies for separating the direct and global components in various fields, such as visualization of the draft layer of a painting [5] and visualization of blood vessels under the skin [3]. However, in most conventional methods, the separation target is captured multiple times under high spatial frequency illumination, and the global component is extracted using digital image processing technologies [2]. Therefore, observers see the separation results not on the surface directly with their naked eyes, but on a computer monitor.

In this study, we constructed an optical see-through device that selectively passes only the global reflection component to an observer's eye by combining high spatial frequency pattern projection and a high spatial frequency mask using a liquid crystal device. The advantage of this method over the previous separation method [1] is that the global component of a specific part of the scene can be shown.

2 THE PROPOSED METHOD

Fig. 1 shows a diagram of the system. A black-and-white high-frequency pattern projection from a projector onto a scene produces bright areas that are directly illuminated by white pixels and dark

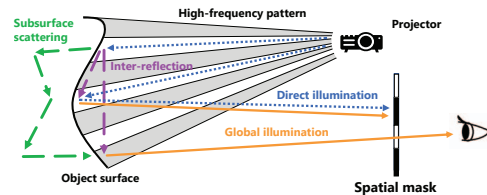


Fig. 1: Conceptual diagram of component separation

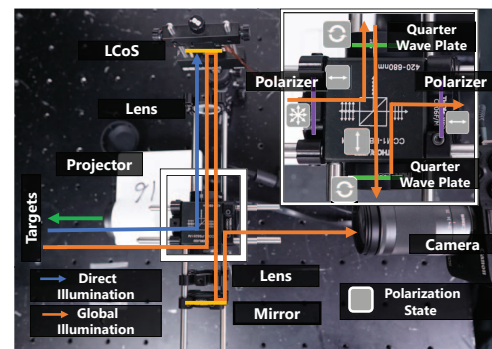


Fig. 2: Configuration of the proposed scope

areas that are not illuminated. The reflected light from the bright areas contains a direct component caused by the direct illumination of a white pixel and a global component caused by reflected or scattered light from other areas. The dark area contains only the global component. Therefore, by observing the dark area alone only the global component is observed.

The proposed system uses a liquid crystal on silicon (LCoS) to selectively block the direct component from the object's surface so that only the dark area can be selectively observed by an observer's eye, thus realizing observation of only the global component. By synchronizing the high spatial frequency patterns of the projector and the LCoS, and by shifting the phase of the patterns faster than the critical flicker fusion (CFF) frequency of the human visual system, the viewer can observe the scene with only the global component without perceiving the patterns.

Fig. 2 shows a prototype of the proposed system. The projector is an Optoma ML1050ST+, and the LCoS is an HED-2220 LCOS. Theoretically, component separation requires two types of patterns: high spatial frequency patterns and their complements. However, more patterns are needed to overcome the optical limitations of the projector. Yet, there is a limit to the number of patterns that can be projected at a higher speed than CFF due to the limitations of the projector and the LCoS frame rate. Therefore, in this experiment, we used the devices' frame rate of 60 Hz and switched four patterns with phase transitions every $\pi/4$ to verify the principle.

3 EVALUATION

We validated the performance of the proposed method by comparing it with the ground truth. The ground truth was calculated using the

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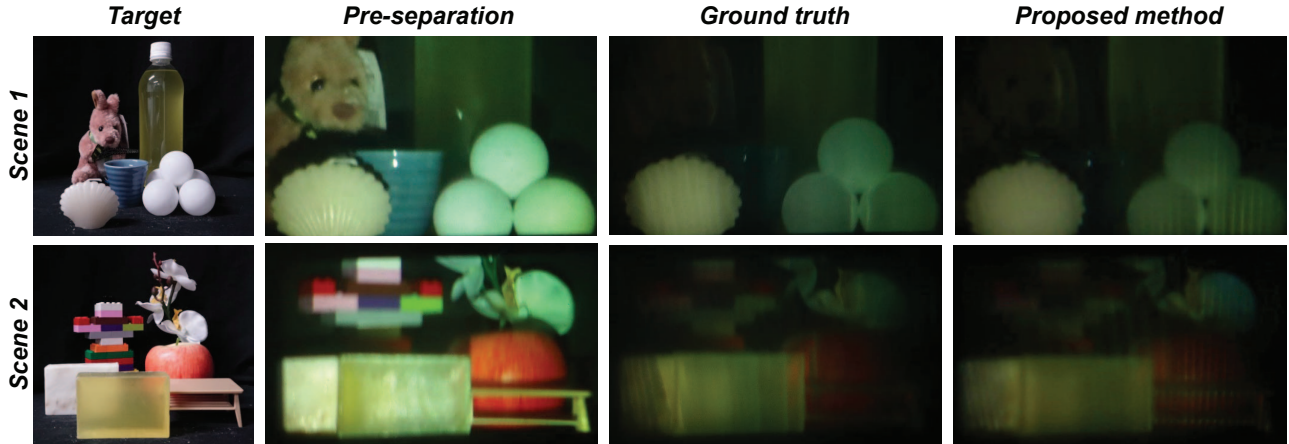


Fig. 5: Experimental results: Pre-separation is the scene from the viewpoint position when the white image is displayed on the projector and the LCoS. Ground truth is the separation result obtained with the conventional method. Proposed method is the separation result obtained with the proposed system.

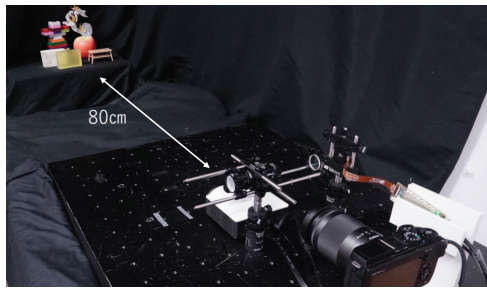


Fig. 3: The experimental setup

conventional method [2], based on digital image processing. Fig. 3 shows the experimental setup. For the ground truth and the proposed method, we used a pattern created using epipolar geometry so that the pixel correspondences between the projector and LCoS did not change depending on the depth of the separation target (Fig. 4).

4 RESULTS

The experimental results are shown in Fig. 5. In the separation results using the conventional method for scene 1, brightness was observed for the juice with Rayleigh scattering, the candle with subsurface scattering, and the ping-pong ball, with inter-reflection between balls. However, no brightened areas were observed in the stuffed toys and ceramics. In the case of scene 2, the brightness was observed for the soap with subsurface scattering and the artificial flowers with inter-reflection between petals. However, no brightness was observed for the other objects.

We compared the separation results of the proposed method with those of the ground truth and found that the separation results were similar for both scene 1 and 2. Therefore, it was confirmed that the proposed method could optically separate components in the same way as the conventional method.

5 CONCLUSION

Reflected light can be separated into a direct illumination component, the primary reflection, and a global illumination component, such as subsurface scattering. Observing only the global component provides a deeper understanding of the reflective properties.

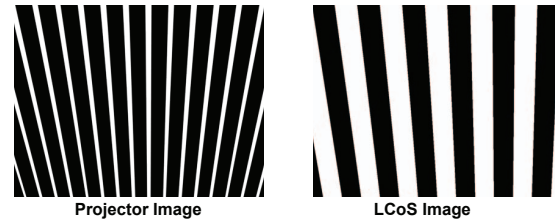


Fig. 4: High-frequency patterns for the projector and the LCoS panel: The patterns projected from the projector are expanded in the dark area to account for calibration errors

In conventional methods, the components are digitally separated electronically using computer vision techniques. In this research, we constructed an optical see-through device that selectively passes only the global reflection component by combining high spatial frequency pattern projection and a high spatial frequency mask. We confirmed that the proposed system could separate components as well as conventional methods. In the future, we will investigate the most accurate pattern for component separation.

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